

THE
BOXING
BOTS

Arduino IDE Setup Guide

macOS Edition — Bluepad32

Introduction

This guide explains how to install Arduino IDE on macOS and configure it specifically for TheBoxingBots kit using the Bluepad32 library. It covers downloading Arduino IDE, installing the correct board package, checking drivers, selecting the correct board and port, and uploading your first program.

Note: Bluepad32 is not a standard Arduino library. It comes as a modified ESP32 board package and must be installed through Boards Manager using a specific URL.

1. Check Your Mac Type

1. Click the Apple menu.
2. Click About This Mac.
3. Check whether your Mac uses Apple Silicon (M1, M2, M3, M4) or an Intel Processor.

You will need this information when downloading Arduino IDE.

2. Download Arduino IDE

1. Open your web browser.
2. Go to: <https://www.arduino.cc/en/software>
3. Download the version that matches your Mac:
 - macOS Apple Silicon — for M1, M2, M3, or M4 Macs
 - macOS Intel — for Intel Macs
4. If a donation screen appears, click Just Download.
5. Wait for the download to complete.

3. Install Arduino IDE

1. Open your Downloads folder.
2. Find the downloaded .dmg file.
3. Double-click the .dmg file.
4. A window will open showing Arduino IDE and the Applications folder.
5. Drag Arduino IDE into the Applications folder.
6. Wait for the copy to finish.
7. Close the installer window.
8. Open the Applications folder.
9. Double-click Arduino IDE to launch it.

4. If macOS Blocks Arduino IDE

If macOS prevents Arduino IDE from opening:

1. Open System Settings.
2. Go to Privacy & Security.
3. Scroll to the Security section.
4. Click Open Anyway next to Arduino IDE.
5. Confirm by clicking Open.
6. Launch Arduino IDE again.

5. Install Xcode Command Line Tools

Some ESP32 compilation tools require Apple's Command Line Tools to be present on your Mac. Install them before setting up any board packages.

1. Open Terminal.
(Press Command + Space, type Terminal, press Enter.)
2. Run this command:

```
xcode-select --install
```
3. If a dialog appears, click Install and wait for it to complete.
4. If the terminal shows:

```
xcode-select: note: Command line tools are already installed.  
— continue to the next step.
```
5. Restart Arduino IDE after installation.

6. Connect Your BoxingBot ESP32 Board

1. Connect the ESP32 board to your Mac using a USB cable.
2. Make sure the board powers on (the power LED should light up).

Note: Some USB cables only charge and do not transfer data. If the board is not detected, try a different USB cable.

7. Check Whether a Driver Is Required

1. Open Arduino IDE.

2. Click Tools → Port.
3. Check whether a serial port appears. Examples:
 - /dev/cu.usbserialxxxx
 - /dev/cu.wchusbserialxxxx
 - /dev/cu.SLAB_USBtoUART

If a port appears, no driver installation is needed. Skip to Step 9.

If no port appears, continue to Step 8.

8. Install the USB-to-Serial Driver

ESP32 boards use a USB-to-Serial chip to communicate with your Mac. If your board is not detected in Tools → Port, install the drivers in this order:

Step 1 — Install the Silicon Labs CP210x Driver

Download the CP210x VCP Driver for macOS from Silicon Labs:

https://www.silabs.com/documents/public/software/Mac_OSX_VCP_Driver.zip

Install the .pkg file. Restart your Mac, reconnect the board, and check Tools → Port again. If a port appears, continue to Step 9.

Step 2 — If the port still does not appear, install the WCH CH340 Driver

Download the CH341SER macOS driver from WCH:

https://www.wch-ic.com/downloads/CH341SER_MAC_ZIP.html

Install the .pkg file. Restart your Mac, reconnect the board, and check Tools → Port again.

Note: You do not need to uninstall one driver before installing the other. Both can coexist on macOS without conflict.

9. Add the Bluepad32 Board Package URL

1. Open Arduino IDE.
2. Go to Arduino IDE → Settings.
3. Find the Additional Boards Manager URLs field.
4. Paste the following into the field:

```
https://raw.githubusercontent.com/ricardoquesada/esp32-arduino-lib-builder/master/bluepad32_files/package_esp32_bluepad32_index.json
```

5. Click OK.

Note: This is the only URL required for TheBoxingBots project. For other boards, open Boards Manager and search directly — most popular boards are already listed there. If a board does not appear, paste its Additional Boards Manager URL into Arduino IDE → Settings the same way as above.

10. Install the Bluepad32 Board Package

1. Go to Tools → Board → Boards Manager.
2. Search for: Bluepad32
3. Install **esp32_bluepad32 by Ricardo Quesada**.
4. Wait for the installation to finish.
5. Restart Arduino IDE.

11. Select the Correct Board

1. Go to Tools → Board → esp32_bluepad32 Arduino.
2. Select **ESP32 Dev Module**.

Important: Always select the board from the esp32_bluepad32 Arduino section, not from the standard ESP32 section. If you select the wrong section, Bluepad32 will not work.

12. Select the Correct Port

1. Go to Tools → Port.
2. Select the port that appears when your board is connected.

On macOS, ports appear as /dev/cu.* (for example: /dev/cu.usbserialxxxx, /dev/cu.SLAB_USBtoUART).

13. Install ESP32Servo Library

1. Go to Sketch → Include Library → Manage Libraries.
2. Search for: ESP32Servo
3. Locate **ESP32Servo by Kevin Harrington**.
4. Click the Version dropdown and select: **3.0.7**
5. Click Install.

Important: Do not install the latest version. Use only ESP32Servo version 3.0.7.

14. Upload the BoxingBot Code

1. Download the BoxingBot code from the link below:

https://ldrv.ms/u/c/01d0e7748238d5f0/IQDX1eLpf748QZP_oBRGhGzkASB2RR5RIg0jObk4CFMCXfY?e=UkQZVt

2. Open the downloaded TBB_FINAL_CODE.ino file in Arduino IDE.
3. Confirm the correct board is selected under Tools → Board (**esp32_bluepad32 → ESP32 Dev Module**).
4. Confirm the correct port is selected under Tools → Port.
5. Click the Upload button (→).
6. Wait for compilation and upload to complete.
7. Arduino IDE will show: **Done uploading**.

15. Troubleshooting

No port appears

- Disconnect and reconnect the board.
- Try a different USB cable.
- Try a different USB port on the Mac.
- Follow Step 8 to install the correct USB driver.
- Restart the Mac.
- Reopen Arduino IDE.

Upload fails

- Confirm the correct board is selected under esp32_bluepad32 Arduino.
- Confirm the correct port is selected.
- Hold the BOOT button while clicking Upload, release when uploading begins.
- Disconnect and reconnect the board, then try again.

Bluepad32 examples not showing in File → Examples

- Confirm esp32_bluepad32 by Ricardo Quesada is installed in Boards Manager.
- Confirm the board selected is from the esp32_bluepad32 Arduino section.
- Restart Arduino IDE completely and check again.

Controller not connecting

- Confirm your ESP32 board has Bluetooth Classic support (standard ESP32 / ESP32-WROOM-32).
- ESP32-S3 and ESP32-C3 do not support Bluetooth Classic and will not work with PS4, PS5, or Xbox controllers.
- Confirm the gamepad is in pairing mode before powering on the BoxingBot.

Arduino IDE will not open

- Open System Settings → Privacy & Security.
- Click Open Anyway next to Arduino IDE.
- Relaunch Arduino IDE.

Mac asks for developer tools during compilation

- Open Terminal and run: `xcode-select --install`
- Install the tools and restart Arduino IDE.